How to get a reproducer of a crash

Use syz-repro to get reproducible program from log

Please use the corresponding version of the Linux kernel and config file.

1. Use syz-execprog to test the effectiveness of reproducible programs

The following are all based on Syz Reproducer to implement local discovery of crashes.

- Compile syz-execprog, syz-executor, syz-exec2prog

Run under syzkaller path:

ompile syz-execprog, syz-executor, syz-exec2prog

```
make execprog
make executor
make prog2c
```

The compilation results are stored in syzkaller/bin/linux_amd64, please note this path.

2. Start the virtual machine

代码块

```
Need a bootable disk image. Syzbot currently uses a small image based on Buildroot which can be built locally or download .NET image.
Local can be used directly with syzkaller's boot image.
```

Disk Image boot:

Disk Image boot:

```
./bin/syz-manager -config=my613.cfg -debug
```

Startup command:

- Note the path of bzImage and bullseye.img

Startup command

```
qemu-system-x86_64 \
  -m 2048 \
  -smp 4 \
  -chardev socket,id=SOCKSYZ,server=on,wait=off,host=localhost,port=58191 \
  -mon chardev=SOCKSYZ,mode=control \
  -display none \
  -serial stdio \
  -no-reboot \
  -name VM-0 \
  -device virtio-rng-pci \
  -enable-kvm \
  -cpu host,migratable=off \
  -device e1000,netdev=net0 \
  -netdev user,id=net0,restrict=on,hostfwd=tcp:127.0.0.1:64317-:22 \
  -hda /path/image/bullseye.img \
  -snapshot \
  -kernel /path/linux-6.14-edit/arch/x86/boot/bzImage \
  -append "root=/dev/sda console=ttyS0"
```

3. Modify virtual machine configuration

- Add new password
- Allow external access

Open the config file:

```
nano /etc/ssh/sshd_config
```

Add the following fields to the file:

```
PermitRootLogin yes
PasswordAuthentication yes
```

Restart the service

```
systemctl restart ssh
systemctl status ssh
```

4. Transfer the reproducer to the virtual machine and run it:

In another terminal, use scp to transfer the compiled repro program to the VM: Use the scp command to transfer the syzkaller binaryall the files in the bin/linux_amd64 directory under the syzkaller directory) and the Reproducer program (the repro.syz file the current directory) to the remote VM (forwarded to the local machine via the SSH port), making sure the port is the same as the one specified in the third steps VM startup command.

Still pay attention to the file path.

Execute the syscall replayer, upload and execute the syscall.

```
scp -P 64317 -o UserKnownHostsFile=/dev/null -o StrictHostKeyChecking=no -o IdentitiesOnly=yes /path/syzkaller/bin/linux_amd64/*  ./100repro.txt root@127.0.0.1:/root/
ssh -p 64317 -o UserKnownHostsFile=/dev/null -o StrictHostKeyChecking=no -o IdentitiesOnly=yes root@127.0.0.1 './syz-execprog -enable=all -repeat=0  -procs=8  -cover=0  100repro.txt'


2. When executing the .C reproducer, you need to upload and execute the compiled result.

# Compile C Reproducer

gcc repro.c -lpthread -static -o repro1
gcc test.c -lpthread -static -o repro1

# Upload the reproduction tool and reproduction program to the virtual machine

scp -P 64317 -o UserKnownHostsFile=/dev/null  -o StrictHostKeyChecking=no -o IdentitiesOnly=yes /path/syzkaller/bin/linux_amd64/* ./repro1 root@127.0.0.1:/root/

#Execution of C Reproducible Program Compilation Results

ssh -p 64317 -o UserKnownHostsFile=/dev/null  -o StrictHostKeyChecking=no -o IdentitiesOnly=yes root@127.0.0.1 'chmod +x ./repro1 && ./repro1'
```